

Explainability of Neural Networks (XAI)

CPSC680: Trustworthy Deep Learning

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Readings

- Readings are updated on the website (syllabus page)
- **Lecture 4 readings:**
 - <https://arxiv.org/abs/1703.01365>
(Integrated Gradients)

Content

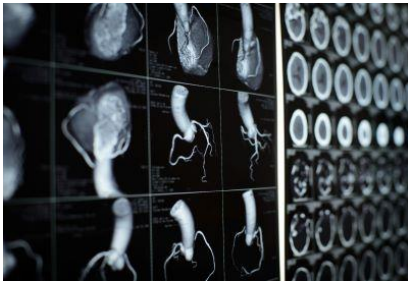
- Introduction to Explainability
- Explainability Settings
- Explainable Models
- Gradient-based Methods
- Perturbation Methods
- (Next lecture: Methods using Surrogate Models)

Explainability

- The **black-box** nature of deep learning makes it a **major challenge** to:
 - Understand what is learned by the ML model
 - Extract insights of the underlying data we are trying to model
- Explainable Artificial Intelligence (XAI) is an umbrella term for any research trying to solve the **black-box problem for AI**
- Why is it useful?
 - Enable users to **understand the decision-making** of the model
 - **Gain trust from human users** of the deep learning system
- Simple-to-read guide: [2004.14545.pdf \(arxiv.org\)](#)

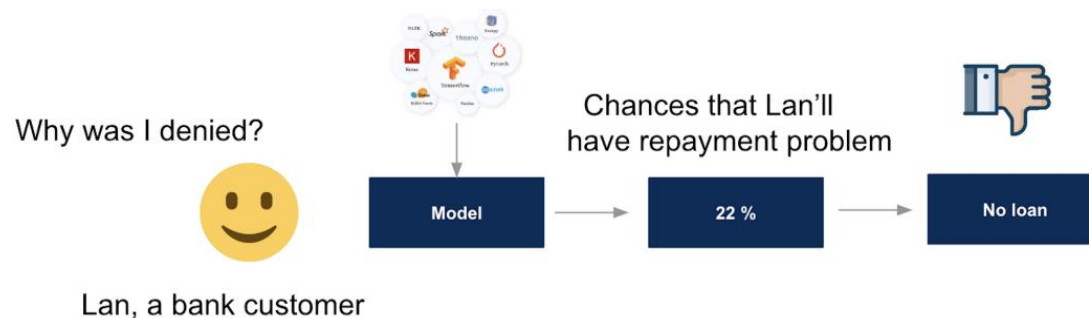
Goal of Explainability

- Model's behavior might be different from the underlying phenomenon
- **Explaining ground truth phenomenon**



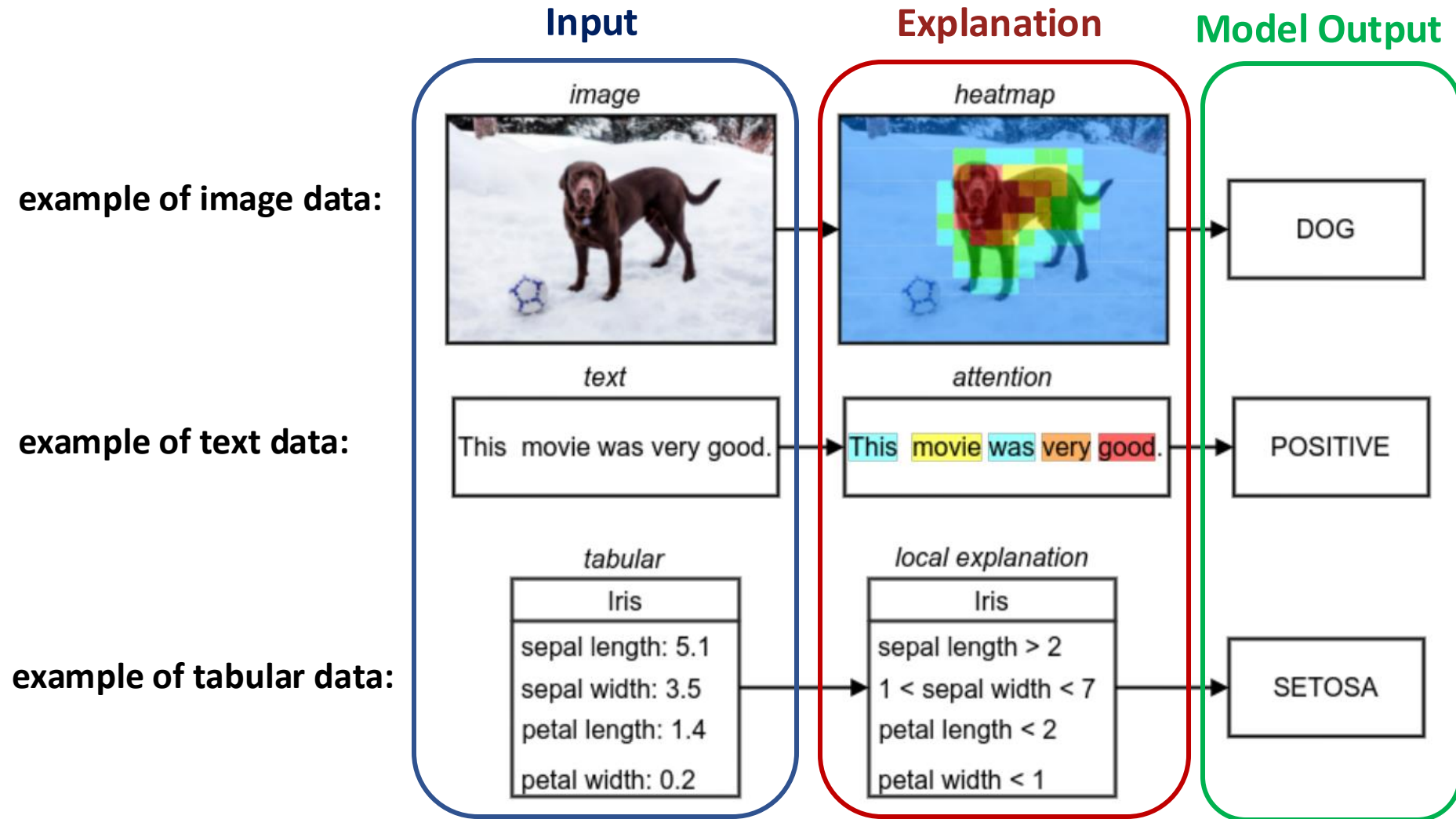
What are the characteristics of certain diseases in terms of imaging?

- **Explaining model predictions**



Why does the model recommend no loan for Person X?

Forms of Explanation

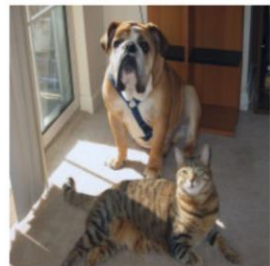


Example: Computer Vision

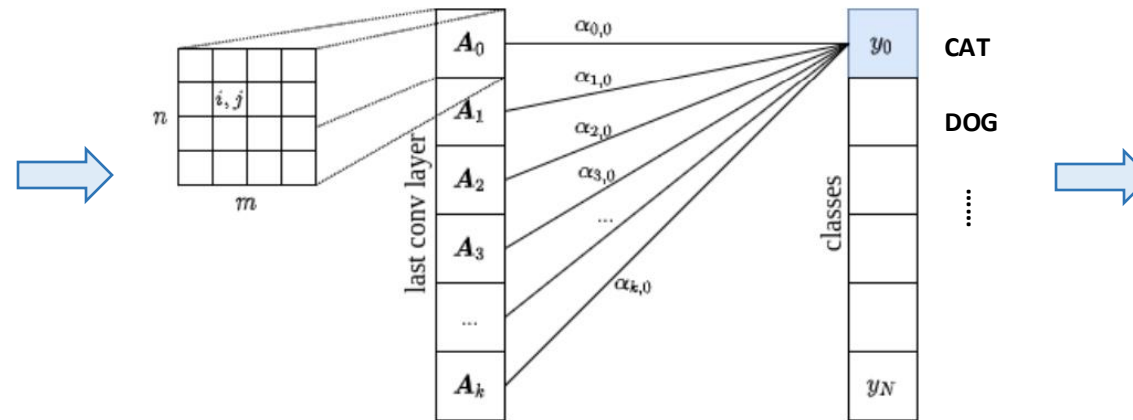
Explanation in Computer Vision:

A particular region of the image **displays the predicted class of objects** (cat / dog in this example)

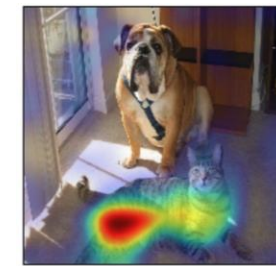
Importance scores on pixels



original graph



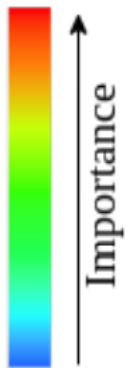
computation process of **CNN** and the prediction



explanation of "cat"

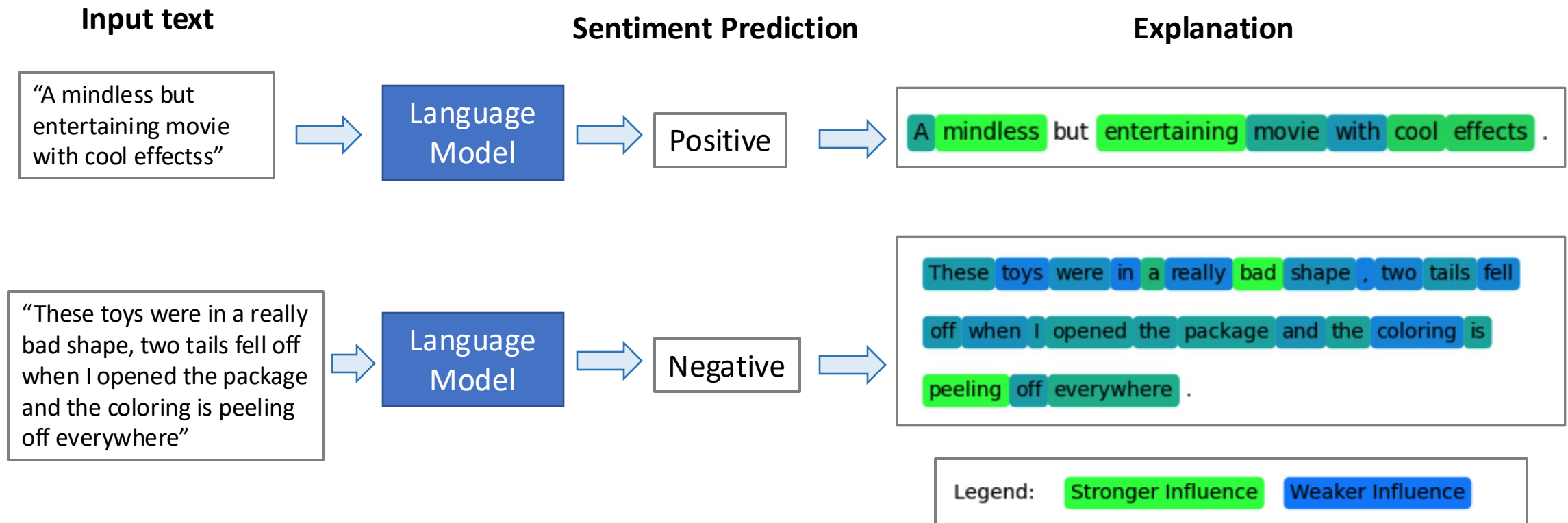


explanation of "dog"



Example: Natural Language Processing

Explanation in Natural Language Processing: important tokens that lead to the prediction

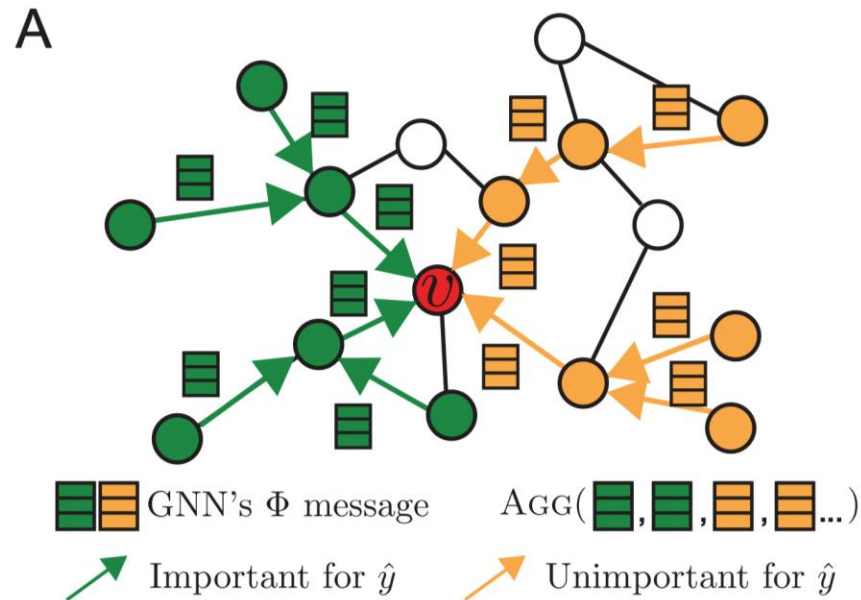


Example: Graph Learning

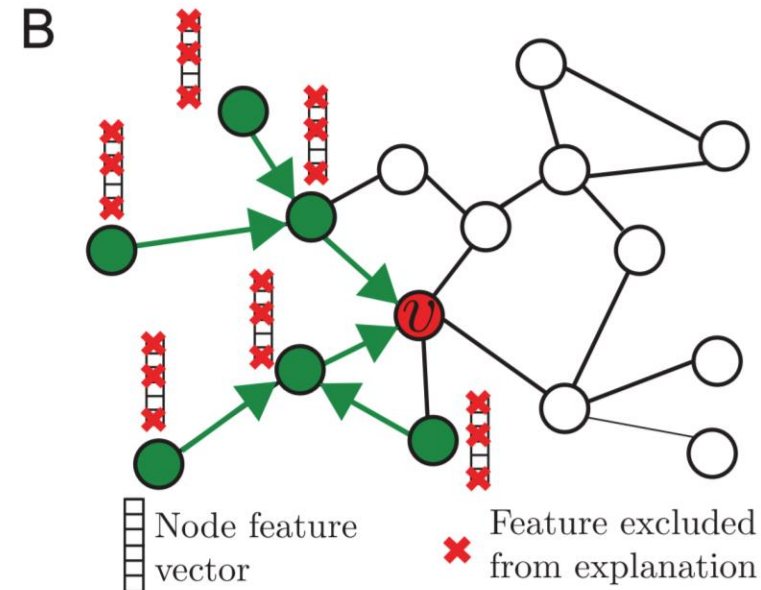
Explanation in Graph Learning: an important **subgraph structure** and a small **subset of node features** that play a crucial role in GNNs prediction

Explanations for prediction at **node v**

A: Import subgraph structure



B: important subset of features



Reasons for Explainability

Why do we need Explainability?

- **Trust:** Explainability is a prerequisite for humans to **trust and accept** the model's prediction.
- **Causality:** Explainability (e.g. attribute importance) conveys **causality** to the system's target prediction: **attribute X causes the data to be Y**
- **Transferability:** The model needs to convey an understanding of decision-making for humans before it can be **safely deployed to unseen data**.
- **Fair and Ethical Decision Making:** Knowing the reasons for a certain decision is a societal need, in order to perceive if the prediction **conforms to ethical standards**.

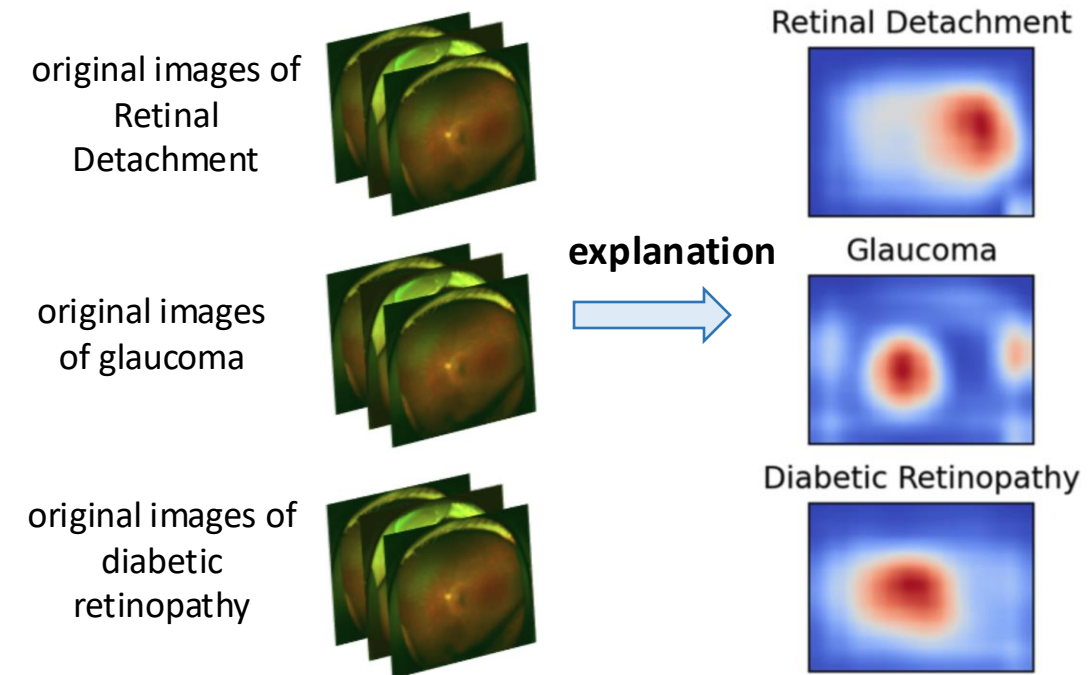
Content

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- **Explainability Settings**
- Explainable Models
- Gradient-based Methods
- Perturbation Methods

Explainability Settings (1)

By target:

- **Instance-level:** a **local** explanation for a single input x and the prediction $\hat{y}$
 - identify the important components of individual instances
- **Model-level:** a **global** explanation for a specific dataset D or classes of D
 - provide **high-level insights** into the model's decision-making behaviors

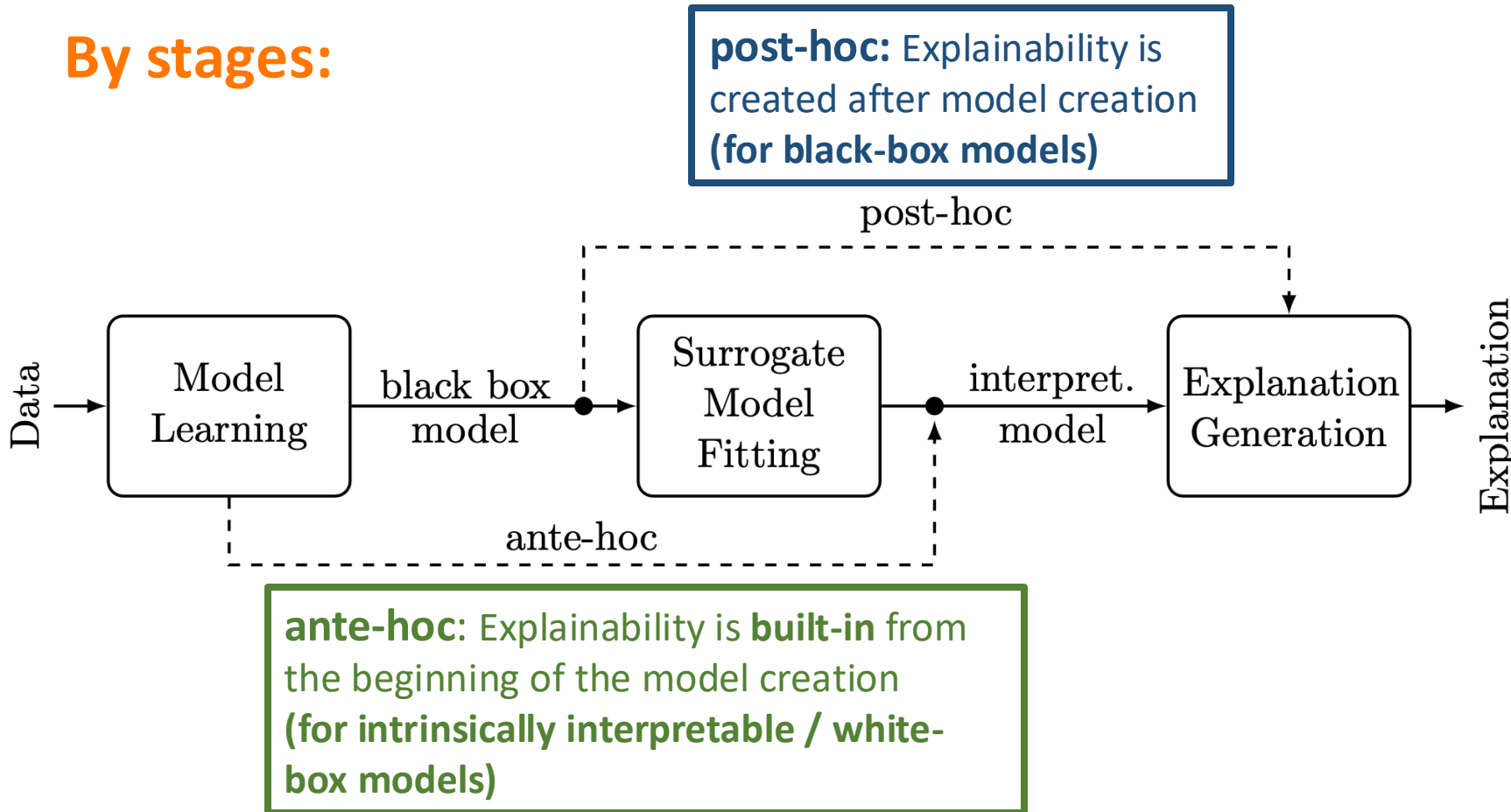


Example: **model-level explanations** for each class

Engelmann, Justin, Amos Storkey, and Miguel O. Bernabeu. "Global explainability in aligned image modalities."

Explainability Settings (2)

By stages:



By applicability of the method:

model-specific: the mechanism for generating explanation is **model-dependent** and works only for a specific model.

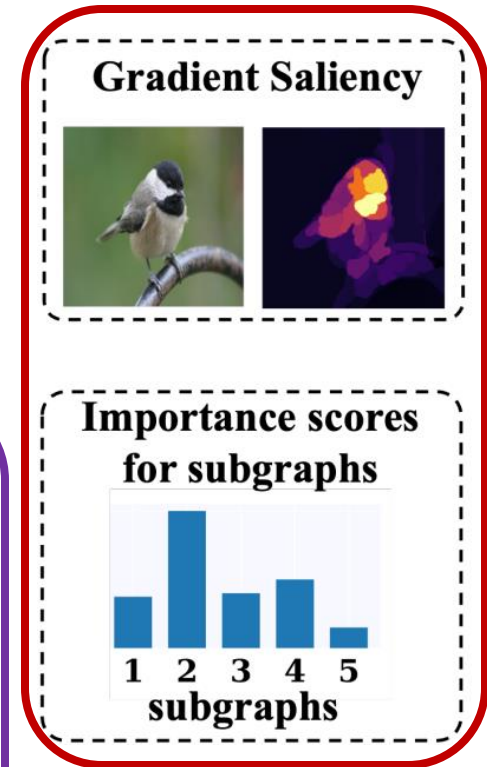
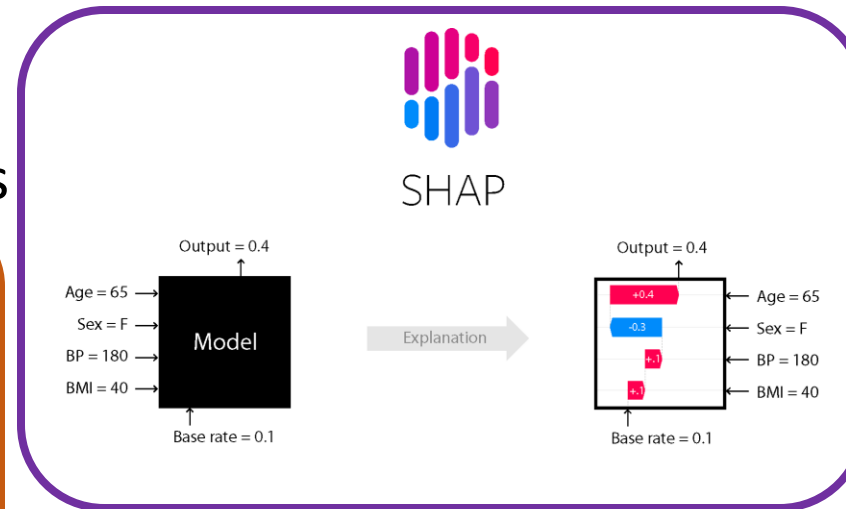
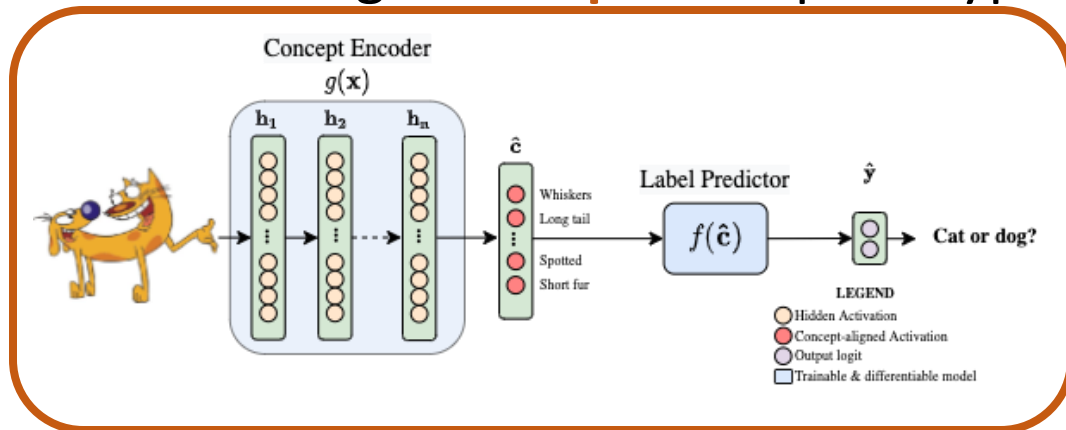
model-agnostic: the mechanism for generating explanation is **applicable** for many or even all model classes

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Explainable Characteristics

- What makes model explainable?
 - **Importance** values (for pixels, features, words, nodes in graphs ...)
 - **Attributions**: straightforward relationships between prediction and input features
 - Encourage **concepts** and prototypes

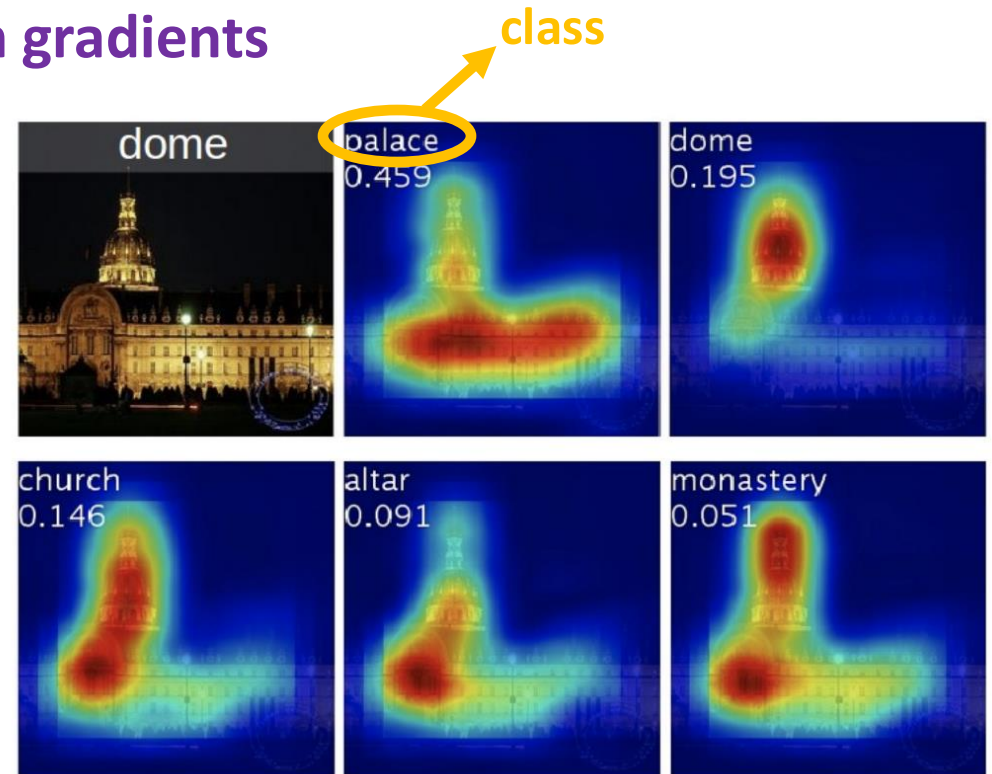


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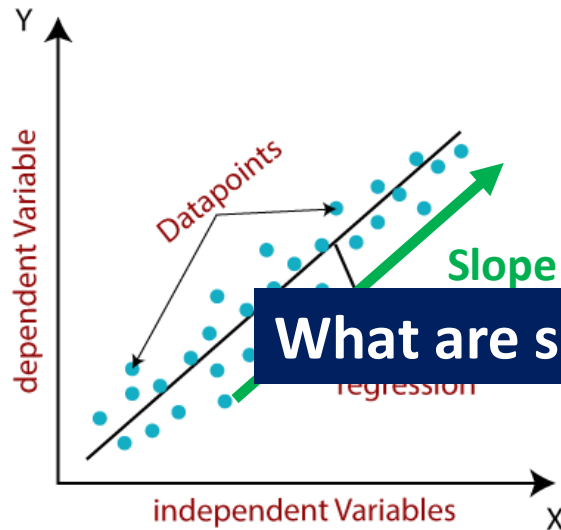
Gradient-based Explanation

- **Gradient-based methods** identify the **saliency of input features** based on gradient signals passed from output to **input features**
- **Intuition: important features tend to have high gradients**
- **We typically care about magnitude**

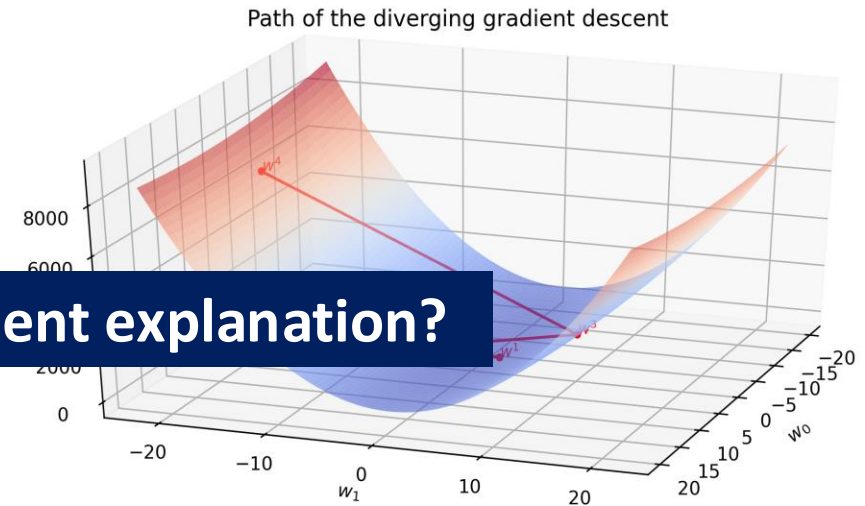


Saliency Map (in the form of a heatmap) highlights the discriminative regions, revealing model's decision-making logic

Why Gradients?



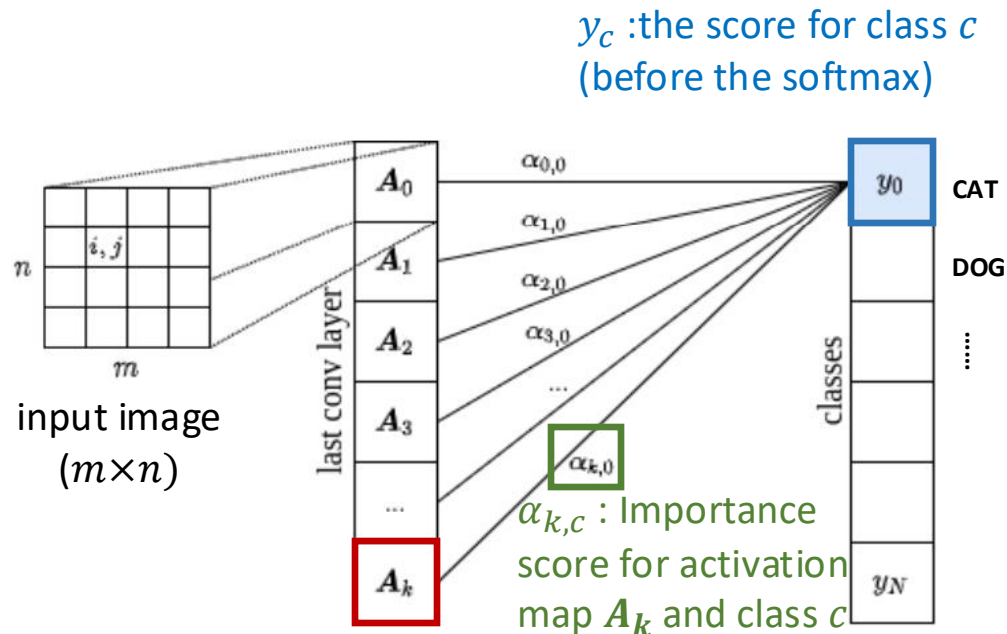
What are some problems with Gradient explanation?



- The **optimization landscape** of deep networks is very complex and a global scope is no longer possible
- Gradient is a **local approximation** of the slope
- Each **dimension** of the gradient vector can indicate how much the prediction is impacted by the input

Grad-CAM (1)

- Gradient-weighted Class Activation Map (Grad-CAM):



Architecture of CNN

gradients of the output w.r.t. the last convolutional layer

Importance score $\alpha_{k,c}$ for activation map A_k and class c :

$$\alpha_{k,c} = \frac{1}{m \cdot n} \sum_{i=1}^m \sum_{j=1}^n \frac{\partial y_c}{\partial A_{k,i,j}}$$

$A_{k,i,j}$: value at (i, j) in the $m \times n$ feature map A_k

Saliency map for class c :

$$\text{map}_c = \text{ReLU}\left(\sum_k \alpha_{k,c} A_k\right)$$

map_c has the same dimension as the input image

Why ReLU?

Grad-CAM (2)

Example: class 0 is “cat”

Saliency map of “cat” :
(of the same size as the
original image)

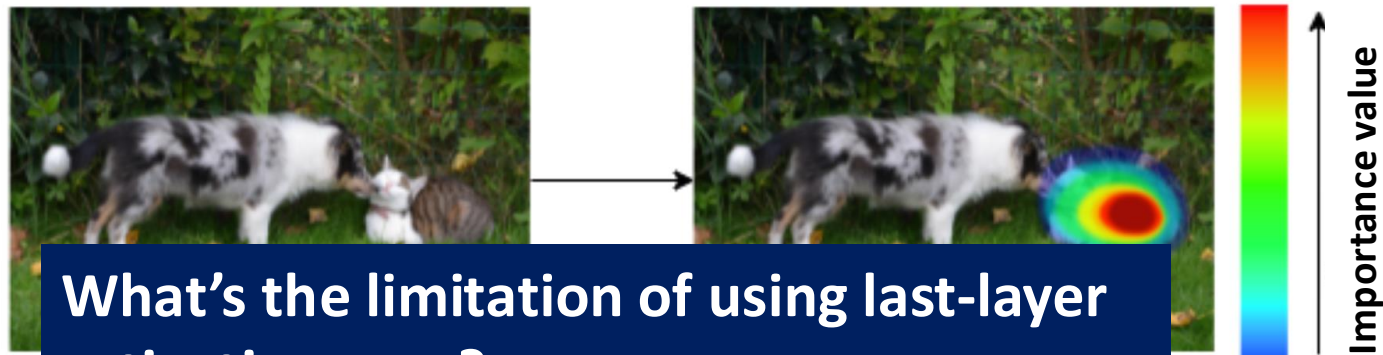
$$map_0 = \text{ReLU}(\alpha_{0,0} A_0 + \alpha_{1,0} A_1 + \dots + \alpha_{k,0} A_k)$$

Important score $\alpha_{k,0} = \frac{1}{m \cdot n} \sum_{i=1}^m \sum_{j=1}^n \frac{\partial y_0}{\partial A_{k,i,j}}$

A_k : feature activation map with size $m \times n$

corresponds to a pixel on the original image

Visualization of the
Saliency map of “cat” :



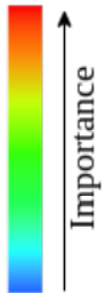
What's the limitation of using last-layer activation map?

Grad-CAM: Evaluation

- **Localization Evaluation:**

- Given an image, first obtain class **predictions from the network**
- Generate Grad-CAM **maps** for **each of the predicted classes**
- **Binarize** with **threshold of 15% of max intensity**

Localize the man and the pizza (ignoring the woman)



Localize the flying kites (despite their small sizes)



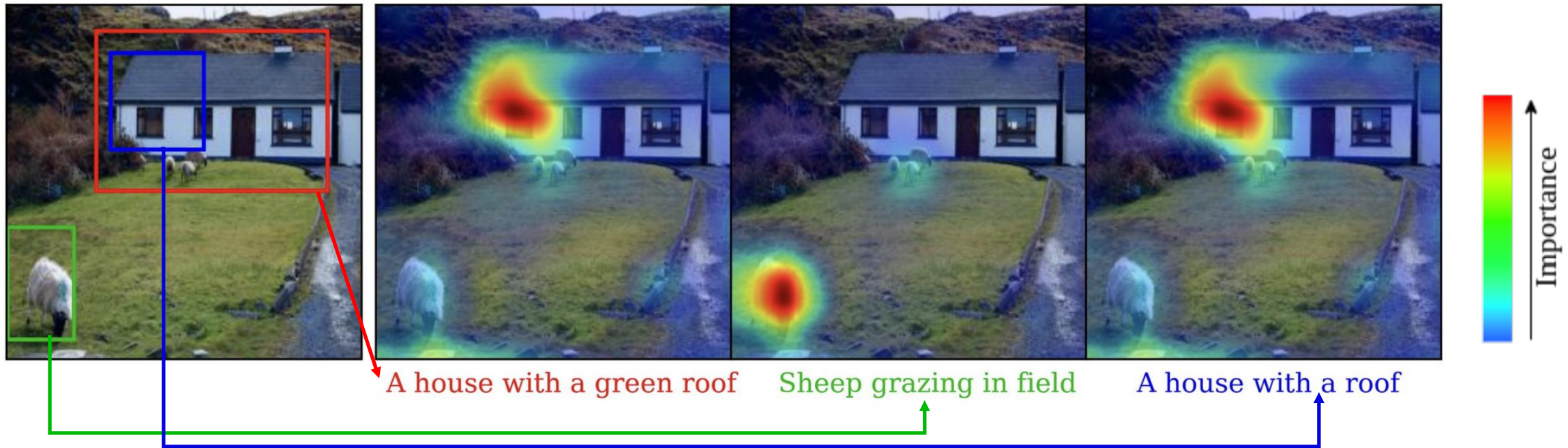
A group of people **flying kites** on a beach

A **man** is sitting at a table with a **pizza**

Visual Explanations highlight image regions that are important for producing the captions

Grad-CAM: Comparison to DenseCap

Localizations of a global caption generated by Grad-CAM:



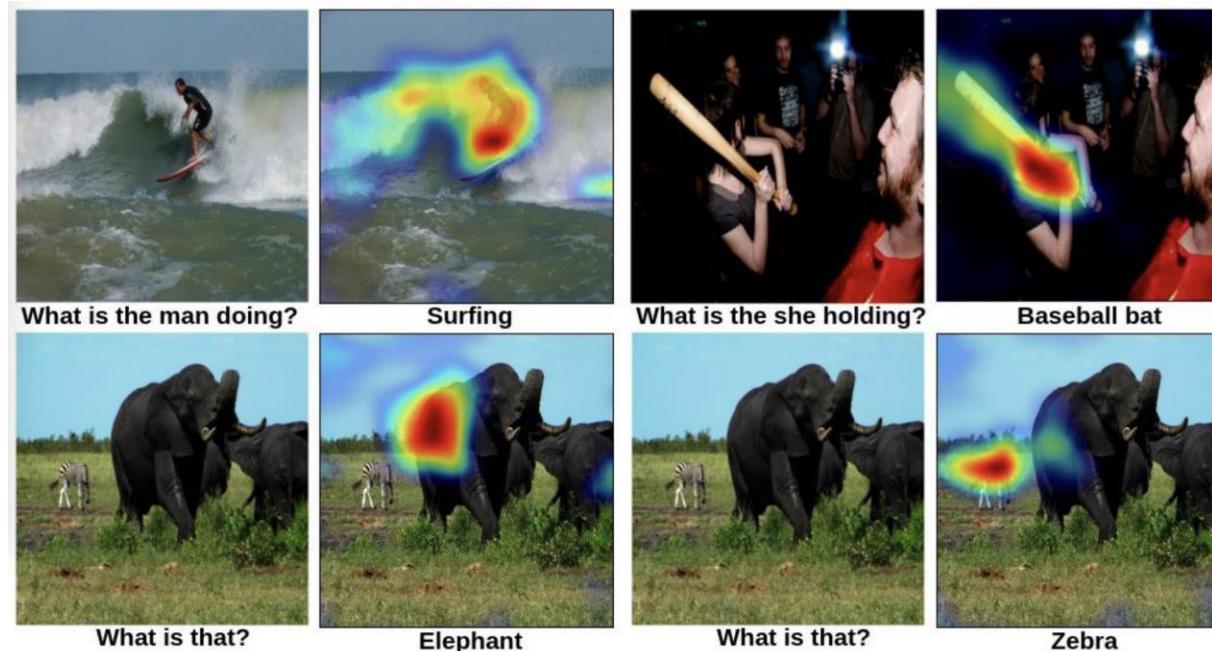
[DenseCap](#): jointly localizes and generates captions for salient regions in each image.

Localization of DenseCap: **bounding boxes** for regions of interest

Localization of Grad-CAM: more **fine-grained** details **with importance values**

Grad-CAM: VQA Evaluation

- **Visual Question Answering (VQA):** VQA pipelines consist of **a CNN to process images** and **a language model for questions**. The model will predict the answer to the question.
- Grad-CAM: **visualize salient regions** of the image that **explain the answer**



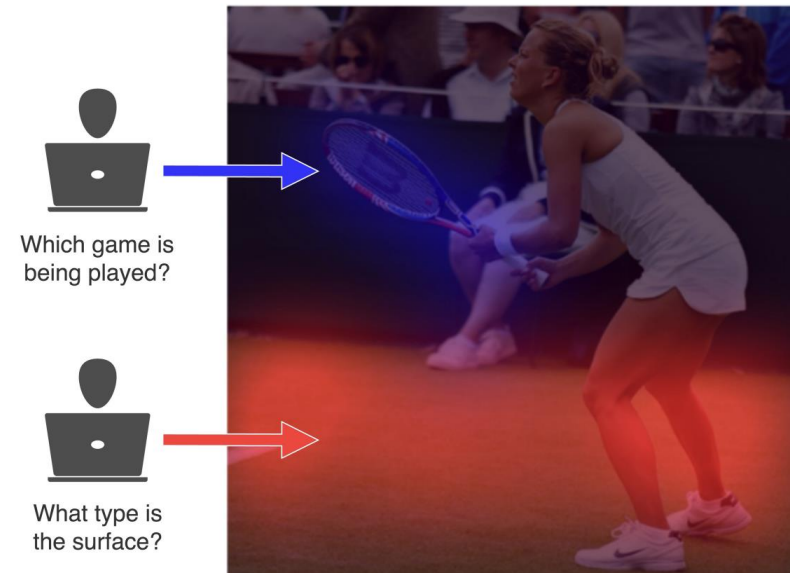
Grad-CAM: Comparison to Human Attention

- Use the **rank correlation** to compare the **Grad-CAM visualizations** and **Human attention maps** over visual question answering pairs
- **Correlation: 0.136**
- statistically higher than chance or random attention maps (**zero correlation**)

Human Attention:

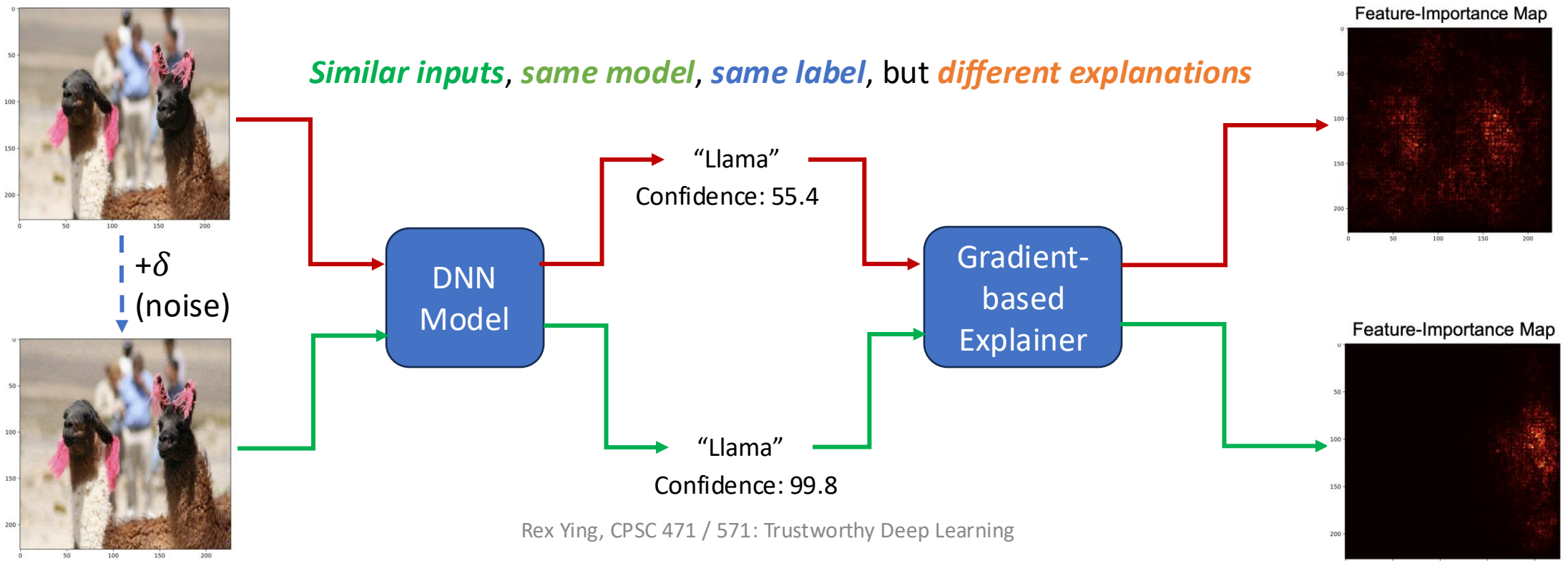
[Das et. al](#) collected human attention maps for a subset of VQA dataset.

These maps have **high intensity** where **humans looked** in the image in order to answer a visual question.



Sensitivity of Gradients

- Saliency maps using a vanilla gradient are sensitive to small perturbations in the input instance.
 - Adding a small perturbation may change the interpretation significantly, even though the prediction is unchanged.



Integrated Gradients (1)

- **Integrated Gradients (IG):** given a **reference (baseline)** input $\mathbf{x}^0$, the integrated gradient of the model f w.r.t. the i -th feature x_i for **an input $\mathbf{x}$** :

$$IG_i(\mathbf{x}) = (x_i - x_i^0) \int_0^1 \frac{\partial f(\mathbf{x}^0 + \alpha(\mathbf{x} - \mathbf{x}^0))}{\partial \gamma(\alpha)_i} \frac{\partial \gamma(\alpha)_i}{\partial \alpha} d\alpha$$

- **$\mathbf{x}$: input**; **$\mathbf{x}^0$: baseline input**; x_i : i -th feature of **$\mathbf{x}$** ; x_i^0 : i -th feature of **$\mathbf{x}^0$**
- **Integral path:** $\gamma(\alpha)$, $\alpha \in [0,1]$

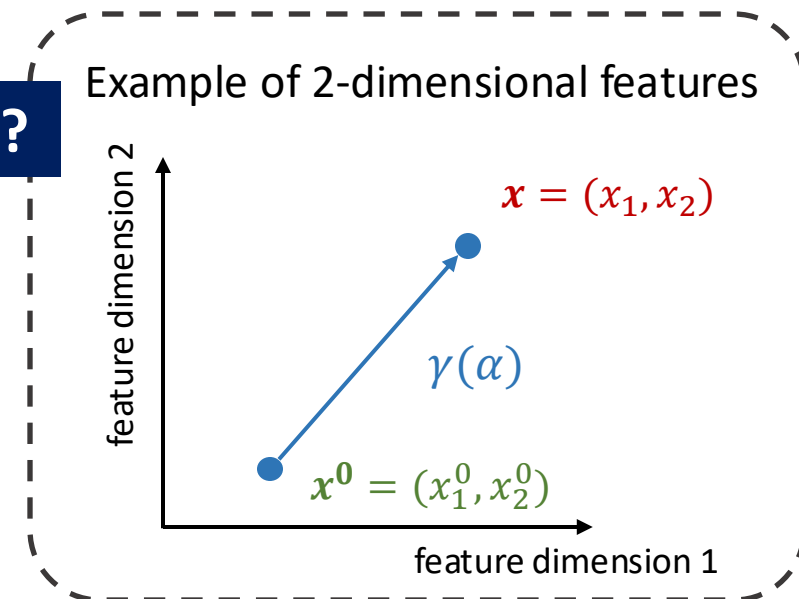
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- $\mathbf{x}$: input; $\mathbf{x}^0$: baseline input; x_i : i -th feature of $\mathbf{x}$; x_i^0 : i -th feature of $\mathbf{x}^0$
- **Integral path:** $\gamma(\alpha) = \mathbf{x}^0 + \alpha \times (\mathbf{x} - \mathbf{x}^0)$, $\alpha \in [0,1]$
- The reference (baseline) input: **Comparison to saliency?**
 - a black image or a zero embedding vector

IG_i : sensitivity of f to changes in the i -th feature from $\mathbf{x}^0$ to $\mathbf{x}$ along $\gamma(\alpha)$ in direction i
Higher $IG_i \iff$ Higher importance of the i -th feature



Integrated Gradients (2)

- IG can be **approximated** by a Riemann summation of the integral

$$IG_i(\mathbf{x}) \approx (x_i - x_i^0) \frac{1}{M} \sum_{k=1}^M \frac{\partial f \left(\mathbf{x}^0 + \frac{k}{M} (\mathbf{x} - \mathbf{x}^0) \right)}{\partial x_i}$$

- M is the number of steps in the Riemann approximation of this integral
(recommended M : 20 to 300 steps)

Observation: Integrated Gradients can better reflect distinctive features of the input image



Top label: reflex camera
Score: 0.993755



Top label: starfish
Score: 0.999992



Integrated Gradients

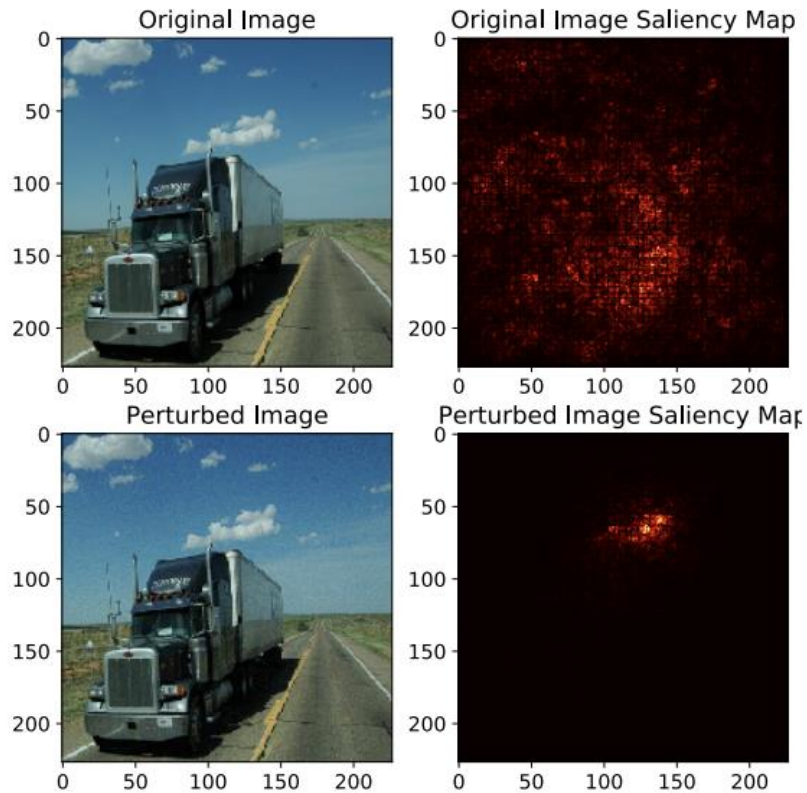
gradients

Original image

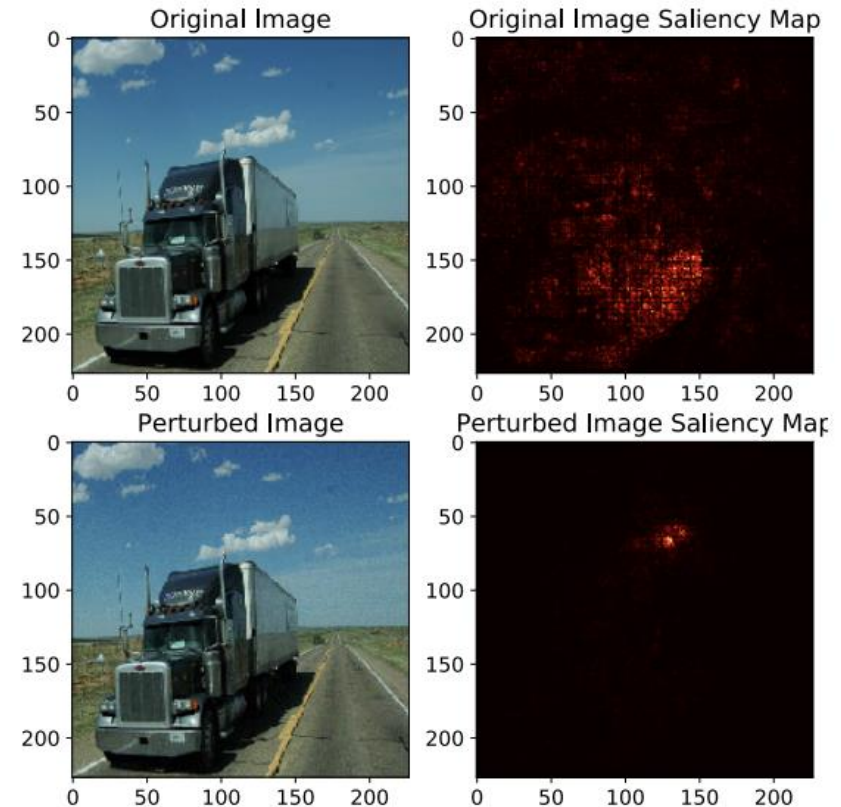
Sensitivity of Integrated Gradients

- Integrated Gradients is still vulnerable to adversarial noise.

Simple Gradient

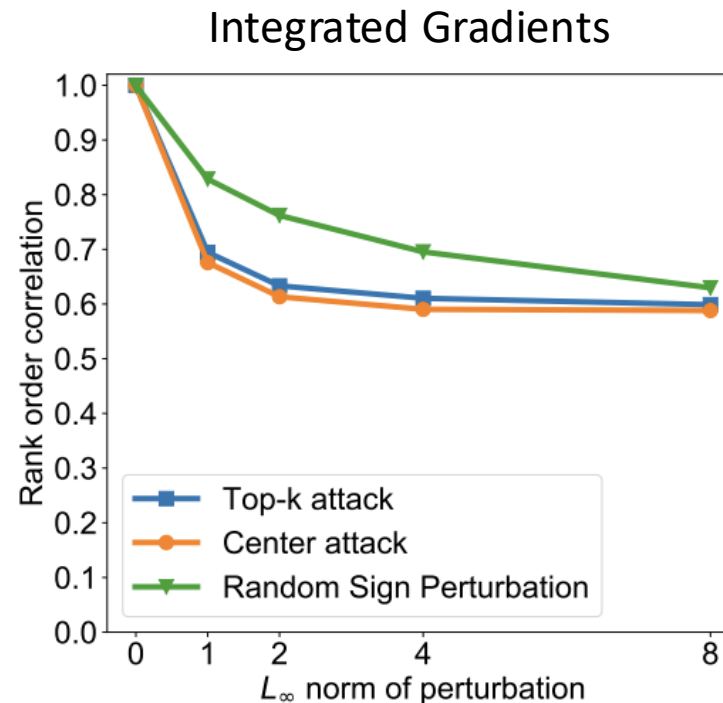
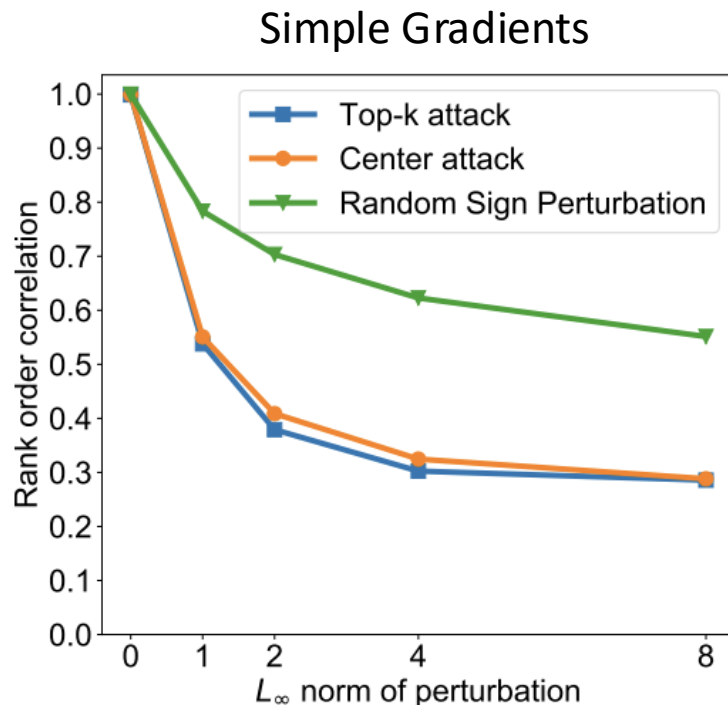


Integrated Gradients



Sensitivity of Integrated Gradients

- Integrated Gradients is still vulnerable to adversarial noise.
- However, statistically Integrated Gradients is **more robust** than the Vanilla Gradient-based method



- **Top-K attack:** optimize the noise to minimize the importance of top-k most important features
- **Center attack:** optimize the noise to maximize the spatial displacement of the saliency map
- **Random sign:** randomly add a random noise $\pm\epsilon$

The y-axis is the correlation of the Saliency map at x_0 and at perturbed input $x_0 + \delta$

Comparison of IG and Grad-CAM

- IG is more **flexible** in the setting than Grad-CAM
 - Adjustable parameters: baseline input, number of steps M , etc.
- IG satisfies the **“sensitivity” atom** by introducing a baseline input
 - **“sensitivity” atom**: if input $\mathbf{x}$ differs from $\mathbf{x}'$ along feature x_i only, and the prediction $f(\mathbf{x}) \neq f(\mathbf{x}')$. Then x_i should have a **non-zero importance score**.
 - Grad-CAM might generate the same saliency maps for $\mathbf{x}$ and $\mathbf{x}'$
- IG is **more robust** to noise / small perturbations on the input
 - The saliency maps of Grad-CAM might change drastically due to small variation of input (e.g., for input $\mathbf{x}$ and perturbed input $\mathbf{x}'$, the prediction $f(\mathbf{x}) = f(\mathbf{x}')$, the saliency maps differ greatly)
- IG has **larger computation complexity**.

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- **When do we need black-box explainability?**
- **How should we tackle black-box explainability?**

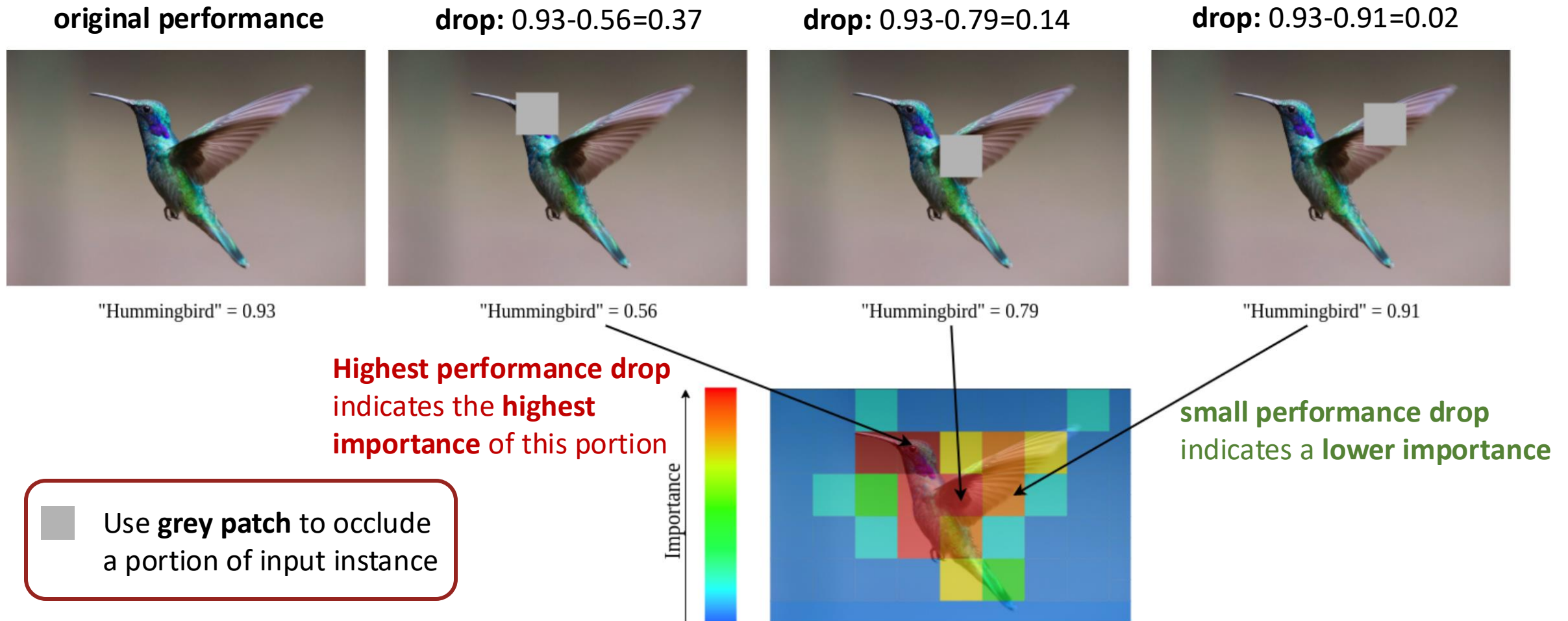
Perturbation-based Explanation

- **Perturbation methods**

- **Post-hoc, model-agnostic** explanation for **black-box models**
- Use perturbation (altering or removing the input features) to identify features that can greatly influence predictions
- **Intuition:** the model's performance **decreases dramatically** when the model does not have access to **the most relevant information**.
- The performance drop can be used to create a **sensitivity heatmap** to visualize the **importance of each portion**

Discussion: How are these techniques related to adversarial attacks?

Principle of Perturbation-based Explanation



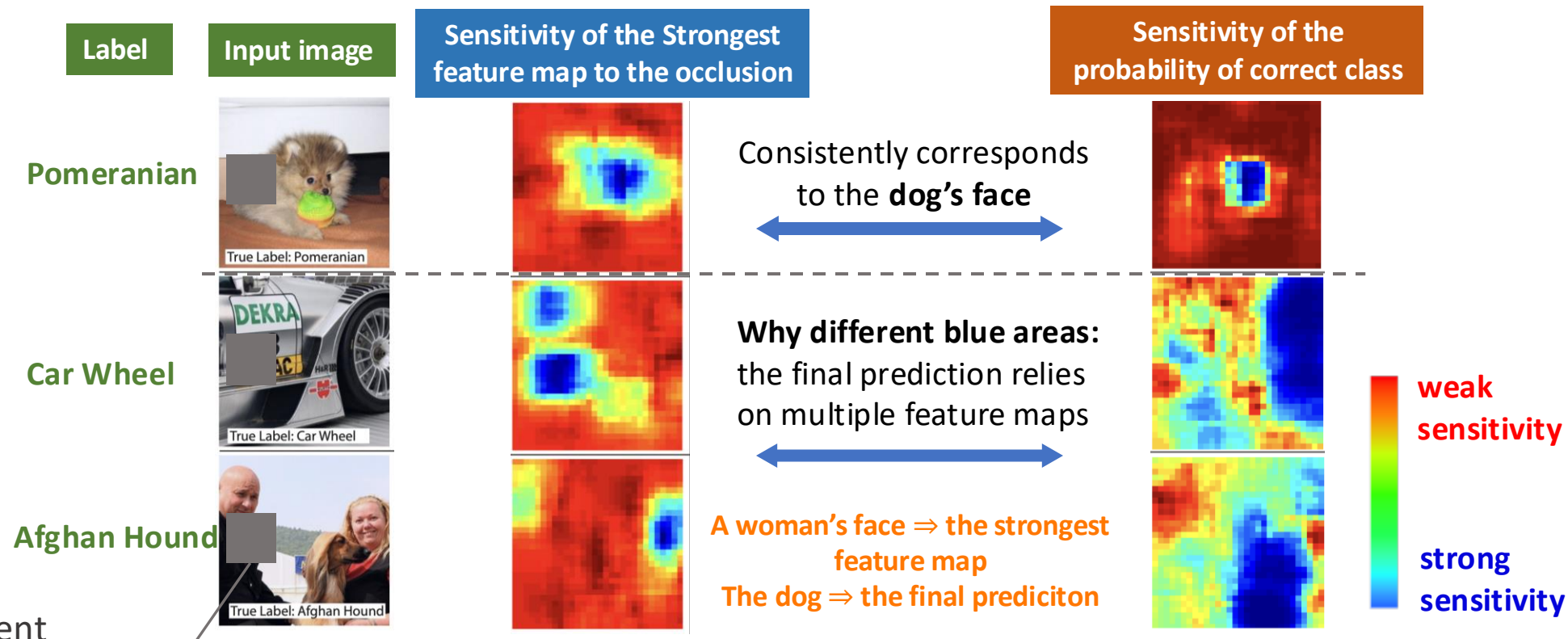
Occlusion Sensitivity

- **Occlusion Sensitivity:** measure the **sensitivity** of the model's output to **occlusion** in different regions by a small grey patch

Strongest feature map: the feature map with the largest values in the top convolution layer

Sensitivity: the change of the value to the occlusion

occlude different portions of the input image with a **grey patch**



Meaningful Perturbation and Evaluation

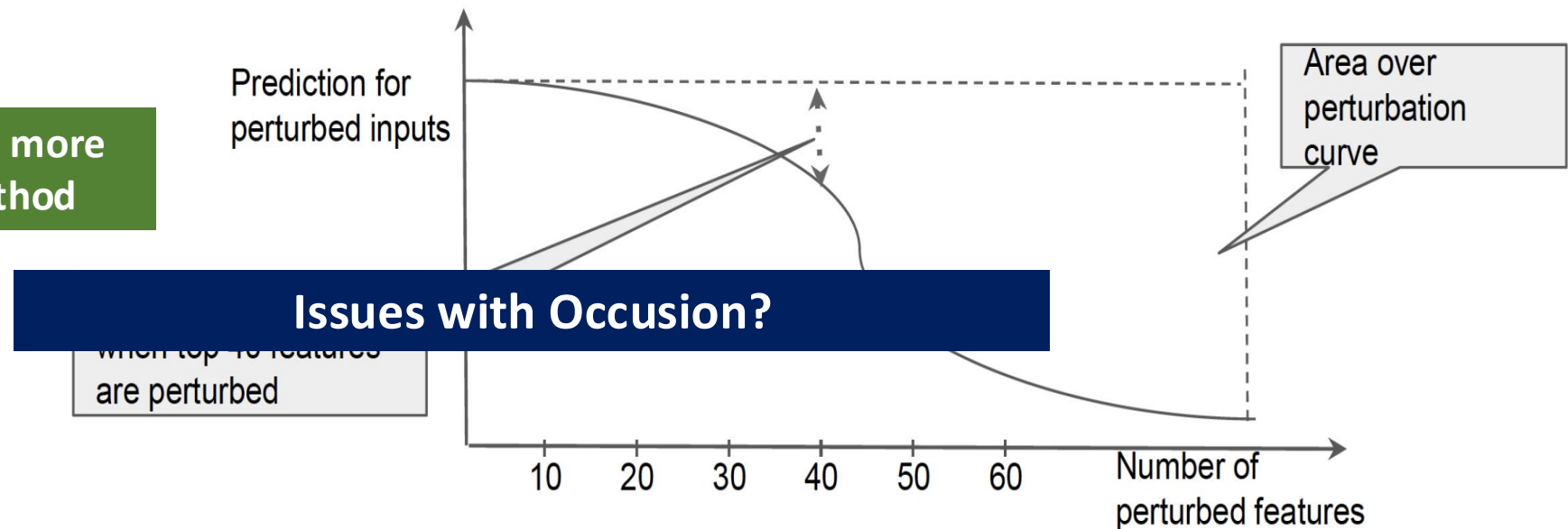
- Other types of **perturbation** [1]:
 - **Blur**: blurring the region area
 - **Constant**: replacing with a constant value
 - **Noise**: adding noise to the region



[1] Fong, Ruth C., and Andrea Vedaldi. "Interpretable explanations of black boxes by meaningful perturbation."

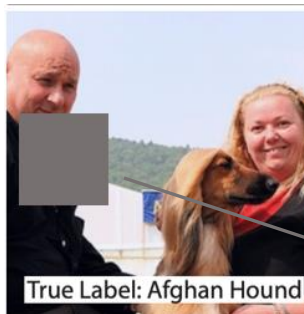
- **Area over Perturbation Curve (AOPC)**: to evaluate the perturbation methods

A higher AOPC indicates a more efficient perturbation method

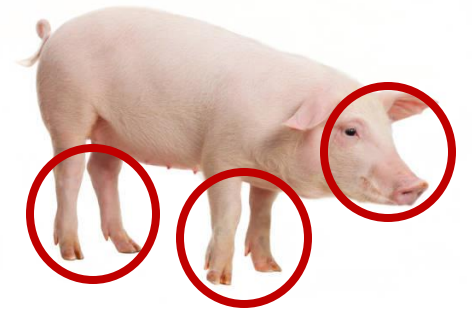


Issues with Occlusion and Perturbation

- Efficiency: occlusion at token / pixel level is very time-consuming
 - Too many forward propagations are needed
- Correlation is not modeled well
 - *E.g.* Presence of Part A “or” part B results in prediction of a class
- The shape of the occluding patch is pre-defined



The size of the **grey patch** is always the same – **same granularity**



Removing one of the patch may not have much effect on the logits

Explanation as Masks

- The learnable mask consists of values between 0 and 1

- $\operatorname{argmin}_M f(\phi(x)) + g(M)$

Masked prediction Regularization

Original input

flute: 0.9973



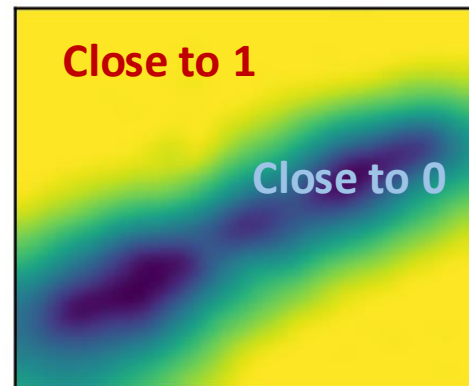
Decrease in model
confidence after
masking

flute: 0.0007



Explanation of
“flute” class in this
instance

Learned Mask



Explanation as Masks

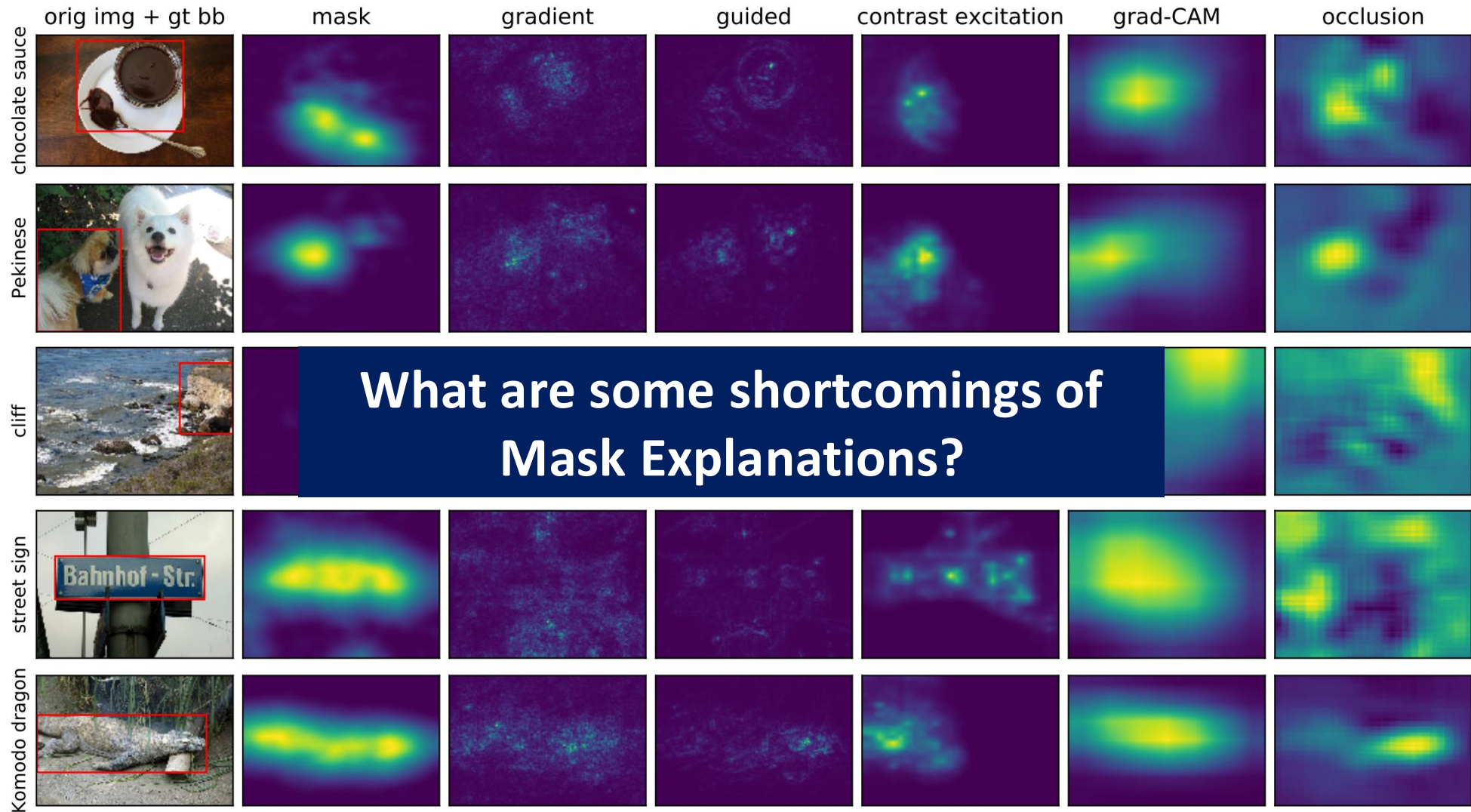
- The learnable mask consists of values between 0 and 1
- $\operatorname{argmin}_M f(\phi(x)) + g(M), \quad 0 \leq M \leq 1$ Mask M is randomly initialized
Masked prediction Regularization

- Replace the masked portion of the input
 - $\phi(x) = x \odot M + x^0 \odot (1 - M)$
- Control the desirable properties of the mask
 - $g(M) = \lambda_1 \|1 - M\| + \sum_u \nabla M(u)$

Image gradient
(difference between
adjacent pixels)

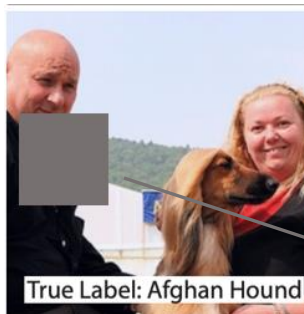
What's the rationale behind $g(M)$?

Comparison with Other Methods

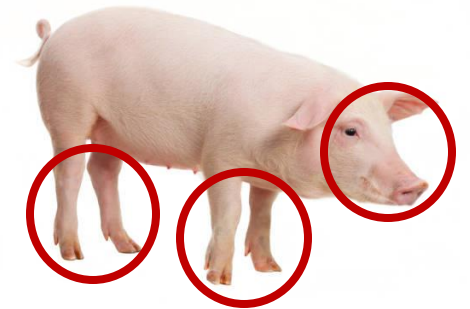


Does Masking Solve These Challenges?

- **Efficiency**: occlusion at token / pixel level is very time-consuming
 - Too many forward propagations are needed
- **Correlation** is not modeled well
 - *E.g.* Presence of Part A “or” part B results in prediction of a class
- The **shape** of the occluding patch is pre-defined



The size of the **grey patch** is always the same – **same granularity**



Removing one of the patch may not have much effect on the logits

What are more issues of masking-based methods?

Problems with Mask Explanations

- **Efficiency**

- Also slow when the input is large (too many pixels, tokens, nodes etc.)
- Requires **optimization** for generating explanation for every instance

- **Stability**

- Explanation can vary across different runs depending on **random seed** of the optimization
- Mask can get stuck in **local optimum** instead of global optimum

- **Robustness**

- The process is analogous to the problem of finding **adversarial examples**
- Explanations might not provide the true insight!

Summary

- The goal of XAI is to enable users to **understand the decision-making** of the model and **gain the trust** of human users of the deep learning system.
- The explanation for a DL system can be categorized as being **model-level or instance-level**; **ante-hoc or post-hoc**; **model-specific or model-agnostic**.
- **Explainable Models** include decision trees, linear models, etc.
- **Gradient-based Explanation**: Saliency, Grad-CAM, Integral gradient
- The change of prediction to **perturbation** over individual regions reveals the importance of the specific region
 - Occlusion and mask-based optimization
- All methods introduced today are instance-level explainability methods